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ECOSYSTEM Technical Group 4

Overview of renewable resources and their availability (Textile Market)

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Executive Summary

This document consists of 2 parts. The first part covers an overview over the different types of renewable materials, with a focus on bio-based, regenerated and recycled. The second part provides an overview over different types of material and their availability on the market as such and in fibre form. This table reflects the current state of the art as identified by the ECOSYSTEM TG4 members and it does not claim to be exhaustive.

The aim is to update this deliverable on a regular basis. If you, the reader identifies any missing material, you are very welcome to contact the ECOSYSTEM TG4 and provide the necessary information for adding this material to the table.

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01

Overview over the types of and sources for renewable resources

Renewable and bio-based terms are quite often used interchangeably, but they present some differences: while renewable refers to what is composed of biomass and can be replenished quickly, bio-based is what is derived from living organisms or their by-products.

By definition of the European Environment Agency, a renewable raw material is «resources that have a natural rate of availability and yield a continual flow of services which may be consumed in any time period without endangering future consumption possibilities as long as current use does not exceed net renewal during the period under consideration.»¹ Bio-based materials are «wholly or partly derived from materials of biological origin (such as plants, animals, enzymes, and microorganisms, including bacteria, fungi and yeast)».² Therefore, while a renewable material is also bio-based, the other way around may not be true.

Wood, bamboo, hemp, and other plant-derived materials, as well as silk, wool, and other animal-derived materials are examples of renewable and bio-based materials, but these last goes beyond and include materials that have undergone a transformation process to give

¹ <https://www.eea.europa.eu/help/glossary/gemet-environmental-thesaurus/renewable-raw-material>

² https://single-market-economy.ec.europa.eu/sectors/biotechnology/bio-based-products_en

them different properties or applications, as are examples packaging pellets made of corn starch, cellulose fibres used in paper or cotton production, and bio-based plastics.

While both renewable and bio-based materials can be sustainable, it is important to consider the entire life cycle of the material, including its production, use, and disposal, when evaluating its sustainability.

Terms and definitions

Throughout this document the following definitions were adopted, as considered the most generally accepted ones:

Bio-based: “derived from biomass”^{3,4}

Bio-based content: “fraction of a product that is derived from biomass”⁴

Biomass: “material of biological origin excluding material embedded in geological formations and/or fossilised”^{3,4} Examples (whole or parts of): plants, trees, algae, marine organisms, micro-organisms, animals.

Biomass content: see bio-based content

Biopolymer: “a polymer comprised, at least in part, of building blocks called monomers, produced from renewable feedstocks such as corn. An alternate definition for biopolymer, includes all biologically produced polymers like DNA, RNA and proteins”⁵

Feedstock: any unprocessed/raw material fed into a manufacturing/conversion process.³

Natural bio-based polymers: “polymers synthesised by living organisms, essentially in the form in which they are finally used.” Examples: polysaccharides, cellulose / starch, proteins, bacterial polyhydroxyalkanoates. After extraction and purification, direct industrial exploitation is possible⁶

Synthetic bio-based polymers: “polymers whose monomers derive from renewable resources but which require a chemical transformation for conversion to a polymer.” Examples: PLA synthesised from corn starch⁶

Renewable: “material that is composed of biomass and that can be continually replenished”⁴

Textile care labelling⁷

GINETEX is the international association for textile care labelling. They define and promote the system of care labelling symbols and coordinates its technical background on an international level. The essential technical elements for its implementation are contained in technical regulations. The care labelling system is maintained to ensure that any new technical and ecological developments together with changes in consumer practices are taken into account.

³ BBI JU annual work plan 2020

⁴ EN 16575:2014 «Bio-based products – Vocabulary»

⁵ BBI JU annual work plan 2020; https://knowledge4policy.ec.europa.eu/glossary-item/biopolymer_en

⁶ <https://www.bpf.co.uk/plastipedia/polymers/polymer-bio-based-degradables.aspx>

⁷ www.ginetex.net

Textile fibre names and labelling

The EU has aligned laws in all EU countries with Regulation (EU) No 1007/2011 on fibre names and related labelling and marking of the fibre composition of textile products, commonly referred to as the Textile Labelling Regulation. This was done to protect consumer interests and eliminate potential obstacles to the proper functioning of the internal market.^{8,9}

Textile exchange has developed an overview over renewable materials¹⁰

Renewable textiles from bio-based polymers and materials

When we consider bio-based textile materials, there are three main sources:

1. Plants
2. Animals
3. Microorganisms

One can distinguish between:

- Fibres derived directly from these sources
- Fibres drawn from regenerated material
- Fibres drawn from bio-based polymeric material

For polymeric material the focus of this document will be on fully bio-based.

It needs to be noted that there are also polymers for which one or more chemical components are bio-based with the remaining still based in petrochemistry. Some examples have been added into the table in section 2. This can be a first step in the process of sourcing all monomers used for synthesising a given polymer from bio-based resources.

Plants

Several plants are being used to produce textile materials. They are a primary source of **cellulose**, the main natural and renewable polymer used to produce textile fibres, both **natural** and **man-made (artificial fibres)**⁹.

Cellulose (natural fibre) include cellulose that occur in nature in fibre shape. The most prominent representative of this group is cotton. Other fibres, like linen and flax are (re)-gaining importance in Europe.

Regenerated cellulose, or man-made cellulose fibres (MMCF) are obtained from dissolving pulp gained from different sources (Wood, agricultural sources, bamboo, old cotton cloth etc.). In these processes, pulp is dissolved and re-shaped/regenerated into fibres appropriate for textile processing. The fibre types are partly defined by the dissolution technology (lyocell viscose, cupro, ionic liquids) and partly by the fibre physical properties (like modal, within the viscose technology)¹¹.

⁸ https://single-market-economy.ec.europa.eu/sectors/textiles-ecosystem/textiles-leather-fur_en#the-textile-labelling-regulation-eu-10072011

⁹ REGULATION (EU) No 1007/2011 OF THE EUROPEAN PARLIAMENT AND OF THE COUNCIL of 27 September 2011

¹⁰ <https://textileexchange.org/materials/>

¹¹

https://www.cirfs.org/file_access/application/files/6316/9036/1027/2023_07_CIRFS_Position_Cellulose_Generic_Fibre_Names.pdf

By legislation the source is not distinguished, but this could be done by marketing. CIRFS advocates for the recognition of "Cellulose" as a broad generic descriptor for all related man-made fibres, akin to the usage of "Rayon" in the United States.

Examples of plants used to produce **natural cellulose-based fibres** are:

- Cotton - from the bolls of the cotton plant (*Gossypium*)
- Kapok - from inside of the kapok fruit (*Ceiba pentandra*)
- Flax or linen - from the bast of the flax plant (*Linum usitatissimum*)
- Hemp - from the bast of hemp (*Cannabis sativa*)
- Jute – from the bast of *Corchorus olitorius* and *Corchorus capsularis*. Can also be from the bast of *Hibiscus cannabinus*, *Hibiscus sabdariffa*, *Abutilon avicennae*, *Urena lobata*, *Urena sinuata*
- Abaca (manila hemp) - from the sheathing leaf of *Musa textilis*
- Alfa – from the leaves of *Stipa tenacissima*
- Coir (coconut) – from the fruit of *Cocos nucifera*
- Broom – from the bast of *Cytisus scoparius* and/or *Spartium junceum*
- Ramie – from the bast of *Boehmeria nivea* and *Boehmeria tenacissima*
- Sisal – from the leaves of *Agave sisalana*
- Sunn – from the bast of *Crotalaria juncea*
- Henequen – from the bast of *Agave fourcroydes*
- Maguey - from the bast of *Agave cantala*
- Kenaf – from the bast of *Hibiscus cannabinus*
- Phormium – from the leaf of *Phormium tenax*
- Pineapple – from the leaf¹²
- Bamboo "linen" – from the stems¹³
- Nettle – from the stalks¹⁴
- Banana stems¹⁵

As to produce regenerated cellulose fibres (artificial fibres), such as (cellulose) acetate, cupro, modal, triacetate, viscose, acrylic, lyocell and modal, the following sources have been used:

- Wood (e.g., eucalyptus, beech)
- Bamboo
- Sugar cane -wastes after extraction of sugar¹⁶
- Seaweeds
- Orange peel
- Coconut shell
- Soy wastes
- Other agricultural sources....
- Waste textiles...

¹² <https://www.cool-organic-clothing.com/pina-fiber.html>

¹³ <https://slowyarn.com/rayon-ingeo-soy-silk-bamboo-and-more/>

¹⁴ www.himalayanwildfibers.com/

¹⁵ <https://www.theguardian.com/sustainable-business/sustainable-fashion-blog/2015/mar/03/wearable-pineapple-banana-fruit-fashion-material>

¹⁶ <https://www.fibre2fashion.com/industry-article/7048/sugarcane-fibres>

The production of nanocellulose fibres, with an increasing exploitation for textile materials, can also be from different renewable sources, such as:¹⁷

- Coconut husk fibre
- Mengkuang leaves (*Pandanus tectorius*)
- Cotton
- *Agave tequilana*
- Barley wastes
- Tomato peels
- Garlic straw residues
- Forest residues
- Corncob residue
- *Gigantochloa scortechinni* bamboo culms
- Industrial waste cotton
- Cassava root bagasse and peelings
- Sugar palm fibres (*Arenga pinnata*)
- Corn straw
- Sago seed shells

Other plants used to produce non-cellulose fibres include:

- (Brown) Seaweeds - Alginate fibre¹⁸
- Corn – Polylactide fibre from the sugar extracted from corn wastes
- Micro-algae

Animals

The **fur of several animals** is also used as renewable sources of fibres to produce textile materials:

- Wool
- Alpaca, llama, camel, cashmere, mohair, angora, vicuna, yak, guanaco, cashgora, beaver, otter 'wool' or 'hair'
- Animal or horsehair from e.g., cattle hair, common goat hair, horsehair

In other cases, such as the silkworms, is their cocoon that is used to extract fibres and not the animal in itself, in this case to produce:

- Silk

Microorganisms

Textile materials from microorganisms is a more recent technology, and there are few examples of resources being employed, such as:

- Fungi, namely mushrooms - to produce fibre networks that resemble leather
- Gram-negative or gram-positive bacteria species, such as *Komagataeibacter*, *Acetobacter*, *Rhizobium*, *Agrobacterium*, *Pseudomonas*, *Salmonella*, *Alcaligenes*, and *Sarcina*, to produce bacterial nanocellulose, which can be feed with varied

¹⁷ Front. Bioeng. Biotechnol., 2021, vol. 9, <https://doi.org/10.3389/fbioe.2021.608826>

¹⁸ Alginate fibres: An overview of the production processes and applications in wound management, 2008, DOI:10.1002/pi.2296

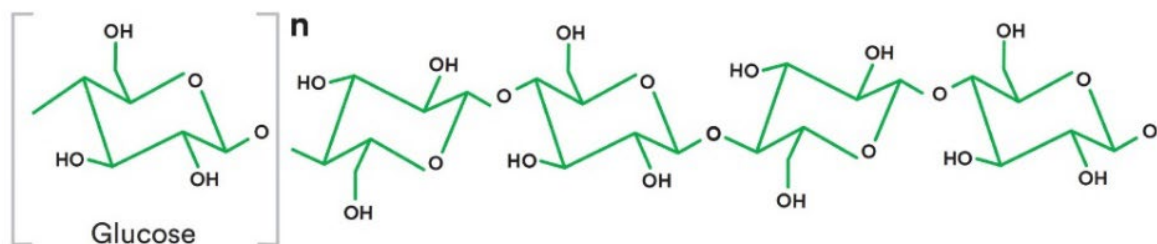
materials, such as sweet tea, soy “wastewater” from the production of Tofu and Tempe and coconut water.¹⁷

Overall, these types of resources offer a more sustainable alternative to non-renewable resources such as petroleum-based polyesters, but we should always keep in mind that the environmental impact of renewable resources depends on the specific processes employed in their production and processing.

Renewable textiles from regenerated fibres

Regenerated cellulose fibres

Cellulose occurs in nature as a polymer of “glucose” units, produced by different plant species. Depending on the source, these polymers show a wide variation in the polymer chain length (Figure 1), which also influences the physical properties of the fibre material.



Average degree of polymerization



Figure 1. The degree of polymerization (polymer chain length) in the main sources of cellulose.

While it occurs already in fibre shape in cotton, hemp and other plants, its other (and larger) occurrences in nature need to be extracted from wood and other plant materials (agricultural waste, bamboo, old cotton cloth, etc.), processed to a dissolving pulp, dissolved and spun again into fibre shape. The result is the so-called regenerated cellulose, or man-made cellulose fibres (MMCF).

The most common MMCF process today is the viscose (rayon) process, in which the dissolving pulp undergoes a chemical reaction, turning it to an alkali-soluble cellulose-xanthogenate. The resulting “viscous” solution is then pressed through nozzles to confer a fibre shape and lead through an acidic bath, where the chemical derivatives are removed, and the fibre is regenerated back to pure cellulose.

Depending on the process conditions, different fibre variations exist besides regular viscose. The most relevant variant for the textile application is the modal fibre, which shows higher tenacity and stability in wet conditions.

Historically, various similar concepts occurred of turning cellulose into something soluble and regenerating as fibre. The most known and still commercially existing type is the cupro fibre, in which the soluble cellulose derivative is a copper complex, and the fibre regenerates back to cellulose in ammonia.

The second generation of regenerated cellulose is based on direct solution (no intermediate chemical derivation). The most common fibre of this group is lyocell, which is produced by dissolving pulp in an aqueous N-Morpholine-N-Oxide (NMMO) mixture, fibre spinning and regeneration on water.

Recently, new dissolving technologies have been developed based on the so-called ionic liquids.

These fibre types are partly defined by the dissolution technology (lyocell, viscose, cupro, ionic liquids) and partly by the fibre physical properties (like modal, within the viscose technology). CIRFS advocates for the recognition of "Cellulose" as a broad generic descriptor for all related man-made fibres, akin to the usage of "Rayon" in the United States. The Regulation that defines the types of fibres that compose a textile material, Regulation (EU) No 1007/2011, is currently under revision by the EC.

While the fibre properties depend on the applied process and its conditions, the resulting fibre itself is independent of the source of the incoming cellulose source (wood pulp, cotton, linen, etc.).

Renewable textiles from recycled polymers and materials

Recycling technologies

Commonly the following main recycling technologies for recycling polymers or textile fibres are:

- Mechanical recycling, resulting in the separation into fibres
- Thermomechanical recycling, resulting in the separation into polymers
- Chemical recycling, resulting in the separation into polymers and monomers
- Thermochemical recycling (cracking), resulting in syngas

A good overview over the current state of the art of recycling technologies can be found in the literature.¹⁹

Availability and quality of recycled material

As in other recycling topics, the availability, homogeneity and purity of the recycling materials is decisive for the feasibility. Cotton is the most present cellulosic fibre in the textile wastes and is therefore the main target (and the most feasible source) for textile recycling. It can be recycled either mechanically by shredding the textiles and processing the re-opened fibre materials into new yarns or batting, or it can be processed as input material to gain soluble pulp, appropriate for chemical recycling into regenerated cellulose fibres.

¹⁹ White Paper on Textile Fibre Recycling Technologies Sustainability 2024, 16(2), 618; <https://doi.org/10.3390/su16020618>

In mechanical fibre recycling, fabrics are disintegrated, as the textile structures must be opened to separate the fibres for reuse in textile processing. The fibre quality coming from mechanical recycling is influenced by many different factors. Of major importance is the source (the waste stream) from which the textile wastes originate (post-industrial, pre-consumer or post-consumer). Another important factor is the construction of the textiles (knitted fabrics, woven fabrics, nonwovens...), which need to be considered in the disintegrated processes. In the fibre mechanical recycling process itself, damage inflicted to the fibres in the ripping and shredding process should be kept as low as possible, so that the proportion of fibres with a suitable fibre length that can be used for spinning into textile yarns is as high as possible.

The MCCF process makes the chemical recycling of cellulosic textiles (pre-dominantly cotton) possible. The involved shredding processes are independent of the fabric types, as a shorter fibre length is even better for further pulping and dissolving processes.

Chemical recycling is the only way to recycle fibre materials too short for mechanical recycling. The most challenging part is removing the non-cellulosic compounds, as a high purity level is required in the existing MMCF processes. Therefore, the cellulose content in the waste stream must be as high as possible to achieve feasible yields and usage of process chemicals. While at the current state of the art feasible technical solutions to separate some synthetic fibres (such as polyester) are available, other synthetic fibres like Elastane and polyamide still represent a challenge.

It must be noted that in chemical recycling, natural fibres such as cotton and hemp, undergo MMCF processes, in which they are turned into regenerated cellulosic fibres, such as lyocell, modal, viscose.

02

Availability of the renewable resources

Currently, the textile industry, as a producer of fashion products for the consumer market and of growing volumes of technical textiles for new areas of application – often replacing conventional materials used so far – ranks among the sectors in which the care about availability of resources must be one of the branch sustainability criteria.

Not only the rising population, but also the extensional consumption of fashion textiles (boosted by “fast fashion” – even though there is an effort to slow down this mode by way of the long lasting “timeless design”) and the expanding application of new technical textiles, cause the steady growth of production of mostly single used fibres. The fibres production has raised by approx. 3% yearly since the 80th of last century, to the actual production of 110 mil ton/year. The prognosis after a slight slowdown during the COVID crisis remains unchanged. Around 65% of the textile consumption is synthetic fibre-based - mostly from limited fossil resources [about 55% polyester (PES) and 5% polyamide (PA)].

Even the around 30% of natural fibres represented by the cotton consumption became somewhat limited, due to: i) climate changes which caused reduction of cultivation areas (e.g., salinity in the Aral Sea area), and ii) the rising range of the alternative, less demanding, plant biomass productions (USA). In fact, 6.5% cellulose fibres originate from wood sources (coming from the responsible forestry certified resources - FSC/PEFC). Other natural fibres (4.5%) include the bast fibre resources. Besides the traditional fibre flax, also the oilseed flax and the hemp fibres are being used – for which yield, and seasonally even quality, can be boosted by enzymatic “bio-retting” (based on the use of customized InoTEX enzymatic products by spraying stalks on the yard or as an additional bath treatment). Recently, there

has been an effort to use other plant fibres (e.g., bamboo, banana tree, pineapple plant, Aloe Vera) for the possibility of complex evaluation of biomass. For the extraction from robust plant structures (decortication), possibilities of support by enzyme processes are again being explored. Thicker fibres are verified mainly for technical purposes and as fibre reinforcement of (bio)composites.

When introducing renewable fibres, it is necessary to consider the currently serious problem of contamination of water and the environment with microplastics (fibre microfibrils from synthetic textiles).

For the EU textile market one of the major threats is the significant dependence on ex-European productions. The inevitable legislative instruments focused on the reduction of waste production will make the whole situation more stringent as well. As today about 80% of textiles end on the landfills or incineration plants. The mentioned legislative instruments will boost the change to the circular economy, for which systematic attention to the use of renewable – natural and bio-based - fibre sources is very important. However, it must be respected that not all bio-based fibres are automatically biodegradable – decomposition takes place under specific optimized conditions.

There are a number of alternative paths to this goal:

- Utilization and revitalization of sources of natural vegetable (bast) fibres including the search for possibilities for the comprehensive wasteless utilization of crop;
- The use of wood (forestry) biomass, or other secondary sources -including waste from agro- and food production, and efforts to introduce emerging - cleaner production through the application of bioprocesses;
- Orientation to rapidly developing industrial biotechnologies for the emergence of new bio-based fibres;
- Develop fully biosynthetic – resulting in biopolymers that offer new processing parameters and final properties, or using bio-based building blocks for their cleaner production instead of the classical polymers used so far. They mostly result in (fibrous) structures with the same properties as classical polymeric materials.

All the overmentioned directions require a multidisciplinary approach, ensuring the attractiveness of the textile industry not only for its new technical staff, but also to achieve the necessary support from other branches (of the agrarian and industrial spheres). Together with the transition from a linear to a circular model of economy, an increase of employment and support for the sustainability of regional and rural activities will be encouraged.

The table below contains some examples of the availability of renewable materials and fibres.

Table 1. Availability of renewable materials used, or with potential use, in the textile industry. **Availability of the material (resource):** High (H): unlimited supply, no problems with getting the material needed; Medium (M): moderate supply, cannot be guaranteed that the needed material is available at all times; Low (L): limited supply, material is available but not sufficient to cover demands; Scarce (S): under development, available on small scale for research purposes. **Availability in fibre form:** Yes (Y); Limited (L): fibres under development; No (N); Unknown (U)

Type of fibre	Origin or Polymer	Renewable source	Availability	Available in fibre form?	Comments
Cellulose	Natural fibre	Cotton	H	Y	
		Linnen	H	Y	
		Hemp	M	Y	Materials available, but research needed to improve fibre quality and properties (coarse yarns). Limited to special products.
		Flax	M	Y	
		Jute	M	Y	
		Kapok	M	Y	
		Nettle	L	Y	
		Pineapple	M	Y	Limited production
		Banana	M	Y	Limited production
		Abaca (Manila fibre)	M	Y	
		Sisal	H	Y	
		Rami	M	Y	
	Bacterial cellulose	Bacteria strains, fed with glucose and fructose rich materials (sugars might come from wastes)	S	Y ²⁰	Very pure, crystalline, can be produced in selected shape
	Regenerated cellulose - Viscose	Dissolving pulp from e.g., wood (eucalyptus, beech, pine, bamboo), other agricultural sources (soy, sugarcane, cotton linters) ²¹	H	Y	The most used manmade cellulosic fibre. Smooth, absorbent, strong, and breathable.
	Regenerated cellulose - Modal	Beech trees ²²	H	Y	
	Regenerated cellulose - Lyocell	Eucalyptus trees ²³	H	Y	
		Seaweed (~19%) + wood pulp	M	Y	SeaCell fibre ²⁴
		Orange pulp	L	Y	Orange Fibre is mixed with wood pulp

²⁰ <https://doi.org/10.3389/fbioe.2020.605374>

²¹ <https://refashion.fr/eco-design/sites/default/files/fichiers/Sustainable%20Material%20Guide%20Viscose.pdf>

²² <https://www.sustainyourstyle.org/en/rayon-viscose-modal>

²³ <https://www.sustainyourstyle.org/en/lyocell-tencel>

²⁴ <https://smartfiber.de/seacell>

Type of fibre	Origin or Polymer	Renewable source	Availability	Available in fibre form?	Comments
					(TENCEL™ lyocell) to produce yarn. ²⁵
	Regenerated cellulose ^(a) - Cupro	Cotton linter ²⁶	L	Y	One of the less well-known man-made cellulosic fibres. Silk like fibre. Currently solely manufactured by Asahi Kasei in Japan under the trademark Bemberg.
	Cellulose acetate	Wood fibres, cotton ²⁷	H	Y	Cellulose fibres where 74-92% of the hydroxyl groups are acetylated
	Cellulose triacetate	Wood fibres, cotton ²⁷	H	Y	Cellulose fibres where at least 92% of the hydroxyl groups are acetylated. High-melting (300 °C), highly crystalline, soluble in limited range of solvents.
Other polysaccharides	Alginate	Alginic acid from brown seaweeds	H	Y	Fibres spinnable only by wet spinning – used mainly in medical textiles (e.g., sutures, wound dressings, tissue engineering)
	Chitin ²⁸	Crab shells, prawn shells, shrimp shells, mushrooms	S	Y	Only R&D. Fibres weak and low tensile strength.
	Chitosan ^{Error! Bookmark not defined.}	Chitosan from deacetylation of chitin from crab shells, prawn shells, shrimp shells or mushrooms	L	Y	Used mainly in medical textiles (e.g., sutures, wound dressings, tissue engineering)
	Chitosan ²⁹ + viscose	Chitosan from deacetylation of chitin from crab shells, prawn shells, shrimp shells or mushrooms + viscose (from wood pulp)	L	Y	Crabyon® fibre.

²⁵ <https://orangefiber.it/process/>

²⁶ <https://www.sustainyourstyle.org/cupro>

²⁷ <https://doi.org/10.1016/j.tifs.2021.06.026>

²⁸ <https://www.diva-portal.org/smash/get/diva2:1444899/FULLTEXT01.pdf>

²⁹ <https://old.swicofil.com/products/055chitosan.html>

Type of fibre	Origin or Polymer	Renewable source	Availability	Available in fibre form?	Comments
	Starch ³⁰	Maize, rice, wheat, potato, cassava, yams, sorghum	S	Y	Only R&D
	Lignin ^{31,32}	Wood pulp, different plants	S	Y	Only R&D
	Thermoplastic starch (TPS) ³³	Starch rich plant e.g., potatoes, rice	S	U	Need to be blended with other polymers
Protein	Natural protein fibres	Wool (from sheep)	H	Y	
		Other animal hair (Cashmere, pashmina, etc.)	M	Y	
		Alpaca	M-H	Y	
		Cashmere	M-H	Y	
		Mohair	M-H	Y	
		Angora	M-H	Y	
		Other animals hair or fur (llama, camel, vicuna, yak, guanaco, cashgora, beaver, otter)	M	Y	
		Animal or horsehair from e.g., cattle hair, common goat hair, horsehair	L-M	Y	
		Silk from silkworm cocoons (with silkworm inside)	H	Y	
		Silk from silkworm cocoons (without silkworm inside)	L	Y	
		Silk from spiders	S	Y	
	Regenerated or synthetic protein ³⁴	Soybean/polyvinyl alcohol	S	U	
		Milk fibres (casein)	L-M	Y	
		Zein from corn Keratin from chicken feather Collagen from leather and hide waste Albumin from eggs	L	Y	Azlon fibre: low in strength when wet and can be stretched to a great extent but does not readily resume its

³⁰ Polymers 2021, 13(7), 1121; <https://doi.org/10.3390/polym13071121>

³¹ <https://www.textileworld.com/textile-world/features/2023/03/carbon-fibers-from-lignin/>

³² Materials (Basel). 2021 Jun; 14(12): 3378, doi: 10.3390/ma14123378

³³ <https://www.packaging.kuraray.eu/blog/thermoplastic-starch/>

³⁴ Produced from either animal or vegetable non-fibrous proteins which have been reconfigured to take up a fibrous form to emulate the natural protein fibres wool or silk)

Type of fibre	Origin or Polymer	Renewable source	Availability	Available in fibre form?	Comments
		Proteins from cotton seeds, peanuts and soybeans			original length. ³⁵ Mainly used in blends.
	Microbial fermentation ³⁶	Plant-based ingredients	M	Y	Brewed Protein™
Bio-based synthetic fibres - Polyesters	PLA	Mostly corn. Other possible sources are sugarcane, tapioca root, cassava, and sugar beet pulp	M	Y	Still limited number of manufacturers; continuity of production not yet guaranteed
	PHB & PHA	PHA copolymers (including PHB) are produced in nature by numerous microorganisms, and through biotechnology	L	L	Development being made to increase range of processability and flexibility, to produce fibres
	Derivatives of PHAs	Chemical modification of biotechnology produced PHBs. Modifiers can be from bio-based source	S	N	Still under R&D
	Polybutylene succinate (PBS)	Succinic acid can be derived from corn starch or sugarcane	L	Y	Low melting point
	Polytrimethylene terephthalate (PTT)	From the two monomers used (1,3-propanediol and terephthalic acid) the 1,3-propanediol can be obtained by fermentation	L	N	
	Polybutylene adipate terephthalate (PBAT)	Partially bio-based (up to 50%) – source unknown	L	N	
	Polyethylene Furanoate (PEF)	The furan dicarboxylic acid can be derived from sugars	L	N	PEF production polymerizes furan dicarboxylic acid and ethylene glycol.
Bio-based synthetic fibres -	Bio-nylon ³⁷	Nylon made from sebacic acid produced from castor	L	L	Only promotional material available. Up to

³⁵ <https://www.britannica.com/technology/azlon>

³⁶ <https://spiber.inc/en/brewedprotein/>

³⁷ <https://www.sportstextiles.toray/en/brand/ecodear/>

Type of fibre	Origin or Polymer	Renewable source	Availability	Available in fibre form?	Comments
Polyamide		oil plants and pentamethylenediamine from corn			10% plant-based. E.g., ecodear™ N510
	Bio PA6.10 ³⁸	Sebacic acid is produced from castor oil plants	L	Y	Up to 63% of the PA is from bio-based source)
	Bio PA11 ³⁹	Castor oil	L	N	High-performance polymer of 100% renewable origin
	Bio PA similar to PA6.6 ⁴⁰	Corn	L	Y	Q-GEO fibre, 46% is bio-based. 50% greater absorbency than a normal PA and greater breathability.
	Bio PA ⁴¹	Castor oil	L	Y	EVO® fibre with quick-drying and odour control properties.
	PA 6, PA 6.6 ⁴²	Tall oil	L	Y	AKROMID® NEXT fibres. Not divulged bio-based content
	PA 6.9 ^{Error! Bookmark not defined.}	Sunflower oil	L	Y	AKROMID® NEXT fibres. 57% bio-based carbon content
	PA 6.10, PA 11 ^{Error! Bookmark not defined.}	Castor oil	L	Y	AKROMID® NEXT fibres. 62-92% bio-based carbon content
	PA 5.6 ^{Error! Bookmark not defined.}	Corn	L	Y	AKROMID® NEXT fibres. 41% bio-based carbon content
	PA 5.10 ^{Error! Bookmark not defined.}	Corn/Castor oil	L	Y	AKROMID® NEXT fibres. 100% bio-based carbon content
	PA610, PA1010 ⁴³	Castor oil	L	Y	VESTAMID® Terra. Up to 100% bio-based
	PA6.6 ⁴⁴	?	L	Y	SENSIL® ByNature. Not divulged bio-based content
	PA4.10 ⁴⁵	?	L	Y	Enka® 70 % bio-based

³⁸ <https://nexisfibers.com/bio-based-pa6-10-fibers/>

³⁹ <https://www.arkema.com/global/en/products/product-finder/product/technicalpolymers/rilsan-family-products/rilsan-pa11/>

⁴⁰ <https://www.fulgar.com/en/products/230/q-geo-by-fulgar>

⁴¹ <https://www.fulgar.com/en/feature/39/bio-based-yarns-evo>

⁴² <https://akro-plastic.com/en/compounds/akromid-next>

⁴³ <https://www.vestamid.com/en/products-services/VESTAMID-terra>

⁴⁴ <https://www.nilit.com/fiber/sustainable/sensil-bynature/?ref=bio-sourced.com>

⁴⁵ <https://mobility.indoramaventures.com/products-markets/new-products-corner/enka-nylon-bio?ref=bio-sourced.com>

Type of fibre	Origin or Polymer	Renewable source	Availability	Available in fibre form?	Comments
	Bio PA ⁴⁶	PA produced using bio-based Pentamethylene diamine (PDA) from renewable plant-based raw materials (not divulged which)	L	Y	TERRY® fibre. 45-100% bio-based carbon content.
Recycled cellulosic materials	Regenerated cellulose ^(a) - Other fibre spinning processes based on ionic liquids	Wood pulp, textiles	L	Y	Still under R&D, in proof-of-concept stage. Ex: Ioncell-F ⁴⁷ Soft and strong fibres even when wet, with high tenacity.
	Recycled cellulose – Renewcell ⁴⁸	Cotton and cellulose rich textile wastes		?	Forms circulose® pulp that may be converted into different man-made cellulosic textile fibres
	Recycled cellulose - Refibra ^{TM49}	Cotton textile wastes + wood pulp		Y	Forms virgin TENCEL fibres
	Recycled cellulose - Evrnu ⁵⁰	Cotton rich textile wastes	L	Y	Output is Nucycl® lyocell fibre. Inherently soft, absorbent, and stronger than virgin cotton and polyester. Commercial manufacturing facility is underway (due in 2025 with annual capacity for 18,000 tons of Nucycl® fibre)
	Recycled cellulose – SaXcell ⁵¹	Cotton textile wastes		Y	Lyocell process
	Recycled cellulose – AeoniQ ^{TM 52}	Different cellulosic raw materials, such as Circulose®, bacterial cellulose and non-valorised agricultural waste	H	Y	Fibres with properties similar to polyester and nylon yarns

⁴⁶ <https://www.cathaybiotech.com/en/brandt.aspx?ref=bio-sourced.com>

⁴⁷ <https://ioncell.fi/>

⁴⁸ <https://www.renewcell.com/en/>

⁴⁹ <https://www.tencel.com/b2b/technologies/refibra-technology>

⁵⁰ <https://www.evrnu.com/>

⁵¹ https://saxcell.com/our_technology.html

⁵² <https://www.heiq-aeoniq.com/>

Type of fibre	Origin or Polymer	Renewable source	Availability	Available in fibre form?	Comments
	Recycled cellulose – Worn Again ⁵³	Textile wastes (PES, PES/CO, PES/cellulose blends with up to 10% other fibres)	L	Y	Outputs are PES pellets and cellulose pulp to produce new fibres. Are implementing 2 nd pilot plant with capacity for 1000 tonnes of textile waste.
	Recycled cellulose – Infinited fibre ⁵⁴	Cellulose rich (at least 88%) wastes (old textiles, used cardboard, crop residues)	L	Y	Output is Infinna fibre: looks and feels like cotton, has antimicrobial properties and superior dye uptake. Known scientifically as cellulose carbamate fibre. Are implementing factory with annual production capacity of 30,000 metric tons.
	Recycled cellulose – Blocktexas ⁵⁵	Textile wastes (PES/CO blends, 100% PES, 100% CO, all cellulosic/man-made cellulosic fibres such as bamboo, flax, rayon, tencel, modal)	L-M	L	Outputs are virgin-quality rPET (PolyTexx®) and microcrystalline cellulose (CellTexx®), to produce new fibres, among others. Are finalising first recycling plant in Australia, with estimated recycle capacity for 4,000 tonnes of textiles in their first year.
	Recycled cellulose – Sodra OnceMore® ⁵⁶	CO/PES textiles wastes (min 70% CO, max 30% PES, in a batch) + wood pulp	H	L	Only white/off-white wastes are accepted. Other cellulosic fibres only accepted at max 5% per batch. Output: OnceMore® pulp that can be transformed into fibres via lyocell or viscose processes.
	Recycled cellulose – mechanical recycling	Textile wastes + virgin cellulose fibres	M-H	Y	

⁵³ <https://wornagain.co.uk/about-us/#technology>

⁵⁴ <https://infinitedfiber.com/about-us/>

⁵⁵ <https://www.blocktexas.com/technology>

⁵⁶ <https://www.sodra.com/en/global/pulp/oncemore/>

Additional information which was not (completely) integrated into the table is available here:

- <https://www.dishafashioninstitute.com/protein-fibre-fabrics>
- <https://win-win.info/sustainable-concepts/biosynthetic-fibres/>

ECOSYSTEMEX

European Community of Practice for a Sustainable Textile Ecosystem

Overview of renewable resources and their availability (Textile Market)

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